

The Count's Square Cookies

Where $j = \sqrt{-1}$ Comes From

November 4, 2026 · Nathaniel

Name: _____ Date: _____

The story

The Count loves squares. Every night he puts cookies on trays in perfect squares, with the same number of cookies in every row and column. *"Ah ah ah!"*

Cookie Monster just wants to eat them. **Oscar the Grouch** just wants to cause trouble. And tonight, Oscar asks The Count for a tray so strange that nobody has ever seen one. To make it, we will need a brand new number called j, and help from **Big Bird**'s best friend, **Mr. Snuffleupagus**. Snuffy for short!

Square trays



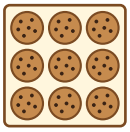
1 row of 1

$1 \times 1 = 1$ cookie



2 rows of 2

$2 \times 2 = 4$ cookies



3 rows of 3

$3 \times 3 = 9$ cookies

Question 1. Fill in The Count's tray table. You can draw the trays or use counters.

Cookies on each side	Times itself	Cookies on tray
1	1×1	1
2	2×2	4
3	3×3	9
4	4×4	
5	5×5	

A number times **itself** is called a **square**, because it makes a square tray. We write 3×3 as 3^2 and say **"three squared."**

Going backwards: the square root

Cookie Monster finds a tray of cookies. He wants to know: *"How many cookies on each side? Me need to know where to start!"*

The **square root** goes backwards. It starts with the cookies on the tray and finds the side. We write it with a check-mark sign:

$$\sqrt{9} \quad \text{means} \quad \text{"what number times itself makes 9?"}$$

$3 \times 3 = 9$, so $\sqrt{9} = 3$. Say it: **"the square root of nine is three."**

Question 2. Use the tray table to go backwards.

$\sqrt{4} =$ _____ $\qquad \sqrt{16} =$ _____ $\qquad \sqrt{25} =$ _____

$\sqrt{1} =$ _____ $\qquad \sqrt{100} =$ _____

Try it! Get 12 counters. Can you make a square tray with all 12, with no counters left over? Now try 16.

Question 3. Which one makes a square tray: 12 or 16? _____

Oscar's surprise

Oscar the Grouch: *"Scram! I know ANOTHER number that times itself makes 9!"*
Remember Oscar's opposite trick: an opposite of an opposite takes you back. So a negative times a negative is **positive**:

$$(-3) \times (-3) = 9.$$

Question 4. $(-4) \times (-4) =$ _____

Question 5. Find **two** numbers that make 25 when you multiply them by themselves.

_____ and _____

So 9 has **two** square roots: 3 and -3. To keep things simple, everyone agreed that the sign $\sqrt{\ }$ means the **plus** one: $\sqrt{9} = +3$. Remember this rule. It comes back at the end!

Oscar's impossible tray

*"Count! Make me a square tray of **minus one** cookie!"* That means we need $\sqrt{-1}$: **a number that times itself makes** -1.

The Count tries every number he knows. *"Ah ah ah! Let me count them!"*

Question 6. Fill in the blanks. Is the answer -1?

$1 \times 1 =$ _____ $\qquad (-1) \times (-1) =$ _____

$2 \times 2 =$ _____ $\qquad (-2) \times (-2) =$ _____

$0 \times 0 =$ _____

Question 7. Did any number times itself make a **minus** number? _____

A plus number times itself is plus. A minus number times itself is plus too (Oscar's trick!). Zero times itself is zero. **No number on the number line works!** We need a number that is *not* on the number line.

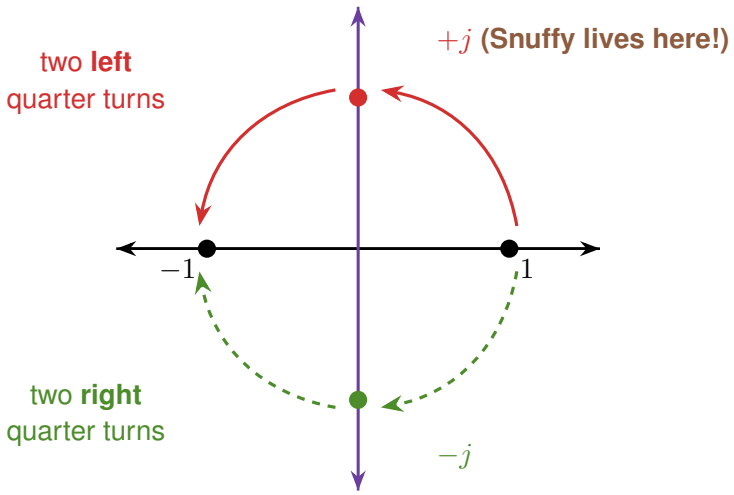
Big Bird: *"Oh! I know someone who isn't on the number line. My friend Snuffy! Nobody ever looks for him there."*

The answer is a turn

Think of **Elmo** standing on the number line, facing the way the numbers grow. Oscar's opposite trick, multiplying by -1, is a **half turn**: Elmo spins around to face the other way, and 1 lands on -1. And "a number times itself" means **doing the same thing two times**. So Oscar's question really asks:

*What turn, done **two times**, makes a half turn?*

Try it! Stand up. Turn **left** a quarter turn, two times. Now start again and turn **right** a quarter turn, two times. Which way are you facing each time?



Meet j, the square root of minus one

A **quarter turn to the left** is the number $\boldsymbol{j}$. It points **up**. Two left quarter turns make a half turn, so

$$j \times j = -1 \qquad \text{and so} \qquad \boldsymbol{j = \sqrt{-1}}\,.$$

Oscar's impossible tray has j cookies on each side! (Engineers, who build radios and phones, call this number j. Mathematicians call it i. Same number, different name.)

Snuffy lives at j, one step **up**, off the number line. For years, the grown-ups on Sesame Street said Snuffy was just Big Bird's **imaginary** friend. But Snuffy was **real** all along! Numbers like j got the same name. A famous mathematician named René Descartes did not believe in them, so he called them **imaginary numbers**. But they are real and useful: engineers use j every day to build radios, phones, and power lines.

Question 8. Elmo turns left twice: $j \times j =$ _____

Question 9. Oscar turns **right** twice. That is $-j$ two times. Which way does he face at the end? _____ So $(-j) \times (-j) =$ _____

So -1 has **two** square roots, $+j$ and $-j$, just like 9 has $+3$ and -3! Remember the rule: the $\sqrt{\ }$ sign means the **plus** one. That is why

$$\sqrt{-1} = +j\,,$$

the left turn that points **up**.

Question 10. Which way does $+j$ point: up or down? _____

The Count's Math Club: write it like an engineer

Scientists and engineers write $j \times j$ as j^2, just like $3 \times 3 = 3^2$:

$$j^2 = -1, \qquad j = +\sqrt{-1}, \qquad (-j)^2 = -1 \text{ too.}$$

Super challenge! $2j$ is two steps up. Check that it is the square root of -4:

$$2j \times 2j = (2 \times 2) \times (j \times j) = 4 \times (-1) = _____$$

Question 11. Use the super challenge to find $\sqrt{-9}$. $\sqrt{-9} =$ _____ j

Draw it!

Draw Snuffy standing at j, one step up. Then draw Elmo making two left quarter turns, from 1 up to Snuffy and over to -1. Write $j \times j = -1$ next to your picture.

For teachers: every nonzero number has two square roots; the radical sign denotes the principal one, which for positive reals is the positive root and for -1 is $+j$ (engineering notation; i in mathematics), by convention the root at $+90°$. The turn picture is multiplication in the complex plane: $\times(-1)$ rotates by $180°$ and $\times(\pm j)$ by $\pm 90°$. Caution for older students: the rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$ fails for negative a and b, since $\sqrt{-1}\sqrt{-1} = j^2 = -1$, not $\sqrt{1} = 1$. Next worksheet: The Cookie Circle. Sesame Street characters appear by name only, for classroom use; this worksheet is not affiliated with or endorsed by Sesame Workshop.

Answer key

Question	Answer
1	4 sides: 16 cookies; 5 sides: 25 cookies
2	$\sqrt{4} = 2$; $\sqrt{16} = 4$; $\sqrt{25} = 5$; $\sqrt{1} = 1$; $\sqrt{100} = 10$
3	16 (a 4 by 4 tray); 12 cannot make a square
4	16
5	5 and -5
6	1, 1, 4, 4, 0; none of them is -1
7	No
8	-1
9	Backwards (toward -1); $(-j) \times (-j) = -1$
10	Up
Super challenge	-4, so $2j$ is a square root of -4
11	$3j$ (because $3j \times 3j = 9 \times (-1) = -9$)